

ETABSPY

ETABSPY is a proprietary software designed to streamline the company's workflow by significantly reducing time spent on repetitive and labor-intensive tasks. It will assist engineers and project engineers in performing various design checks and provide enhanced reporting capabilities within a single application. By eliminating the need for multiple Excel sheets to convert forces or reactions from an ETABS model to S-CONCRETE, ETABSPY will improve efficiency and consistency across projects.

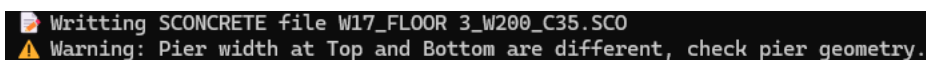




2 Use Cases and Implementation Strategy

1. Model Health Check

ETABSPY will perform an automated health check of the ETABS model. Key aspects such as geometry, load combinations, and element assignments will be reviewed, and a concise summary report will be generated to ensure the model is ready for further analysis.



```
Writting SCONCRETE file W17_FLOOR 3_W200_C35.SCO
Warning: Pier width at Top and Bottom are different, check pier geometry.
```

2. Seismic Irregularity Detection

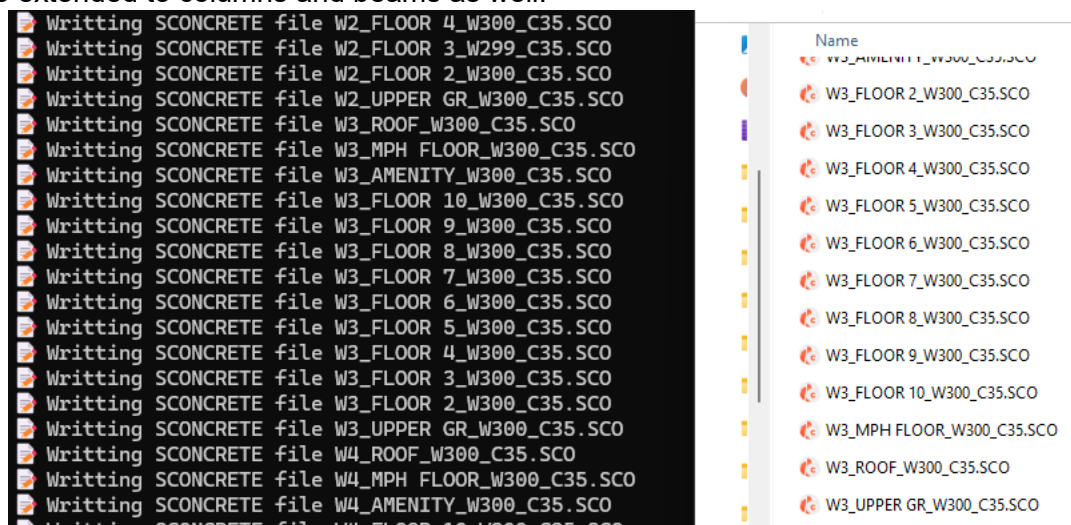
The software will identify seismic irregularities in accordance with applicable code provisions, helping engineers assess compliance and flag potential issues early in the design process.

3. Preliminary Shear Wall Design

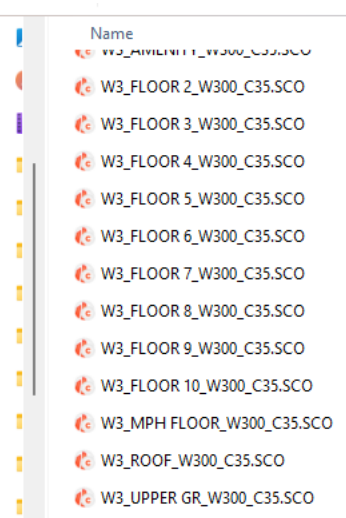
ETABSPY will evaluate all shear walls in the model to determine which can adequately resist lateral loads with minimal code reinforcement. This allows engineers to rule out several elements before proceeding to detailed design in S-CONCRETE.

4. SCONCRETE File Generation

For walls requiring detailed design, the software will automatically generate SCONCRETE files based on user-defined parameters. It will leverage S-CONCRETE's batch processing capabilities to run designs and compile a comprehensive wall schedule as well. This capability can be extended to columns and beams as well.



```
Writting SCONCRETE file W2_FLOOR 4_W300_C35.SCO
Writting SCONCRETE file W2_FLOOR 3_W299_C35.SCO
Writting SCONCRETE file W2_FLOOR 2_W300_C35.SCO
Writting SCONCRETE file W2_UPPER GR_W300_C35.SCO
Writting SCONCRETE file W3_ROOF_W300_C35.SCO
Writting SCONCRETE file W3_MPH FLOOR_W300_C35.SCO
Writting SCONCRETE file W3_AMENITY_W300_C35.SCO
Writting SCONCRETE file W3_FLOOR 10_W300_C35.SCO
Writting SCONCRETE file W3_FLOOR 9_W300_C35.SCO
Writting SCONCRETE file W3_FLOOR 8_W300_C35.SCO
Writting SCONCRETE file W3_FLOOR 7_W300_C35.SCO
Writting SCONCRETE file W3_FLOOR 6_W300_C35.SCO
Writting SCONCRETE file W3_FLOOR 5_W300_C35.SCO
Writting SCONCRETE file W3_FLOOR 4_W300_C35.SCO
Writting SCONCRETE file W3_FLOOR 3_W300_C35.SCO
Writting SCONCRETE file W3_FLOOR 2_W300_C35.SCO
Writting SCONCRETE file W3_UPPER GR_W300_C35.SCO
Writting SCONCRETE file W4_ROOF_W300_C35.SCO
Writting SCONCRETE file W4_MPH FLOOR_W300_C35.SCO
Writting SCONCRETE file W4_AMENITY_W300_C35.SCO
Writting SCONCRETE file W4_FLOOR 10_W300_C35.SCO
```



Name
W3_AMENITY_W300_C35.SCO
W3_FLOOR 2_W300_C35.SCO
W3_FLOOR 3_W300_C35.SCO
W3_FLOOR 4_W300_C35.SCO
W3_FLOOR 5_W300_C35.SCO
W3_FLOOR 6_W300_C35.SCO
W3_FLOOR 7_W300_C35.SCO
W3_FLOOR 8_W300_C35.SCO
W3_FLOOR 9_W300_C35.SCO
W3_FLOOR 10_W300_C35.SCO
W3_MPH FLOOR_W300_C35.SCO
W3_ROOF_W300_C35.SCO
W3_UPPER GR_W300_C35.SCO

5. SAFE File Generation

ETABSPY will streamline the export of ETABS .e2k floor models to SAFE by automating the creation of SAFE files, including the definition of load combinations and load sets. This automation eliminates the need for repetitive manual setup—a task that is currently time-consuming and error-prone. The benefit becomes especially significant during revisions and minor design modifications, where small changes requested by stakeholders can result in disproportionately large overhead due to the manual rework involved.



6. **Automated Rebar Design Reports**

The tool will extract results from SAFE and generate clear, structured rebar design reports for slabs and foundations, reducing time spent interpreting raw output.

7. **Punching Shear Checks**

Automated punching shear checks will be performed for all applicable elements, flagging overstressed zones and suggesting corrective measures where necessary.

8. **Element Change Tracking and Impact Analysis**

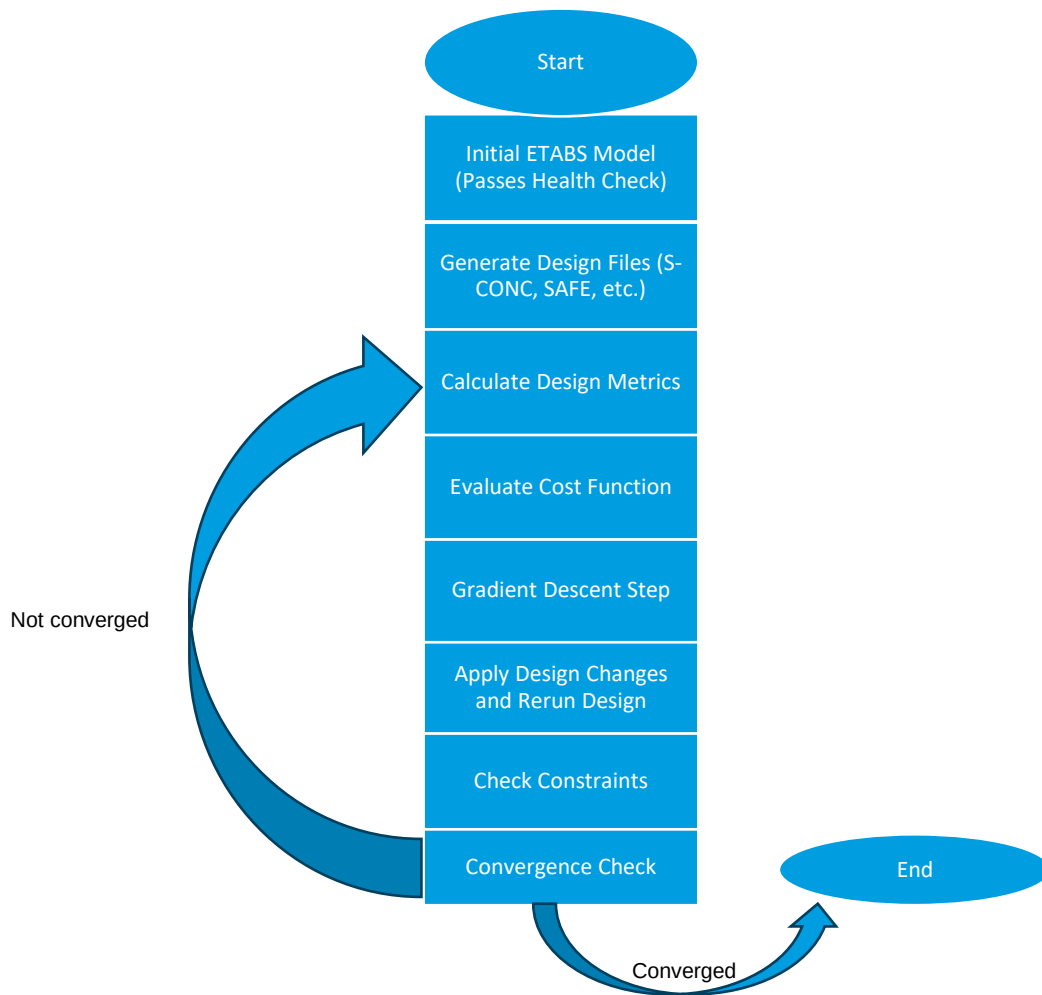
When structural elements such as columns are moved or modified, ETABSPY will track the changes and highlight affected areas of the model, guiding designers to all relevant updates that may be required.

3 Possible Future Additions

1. **Design Optimization Tools**

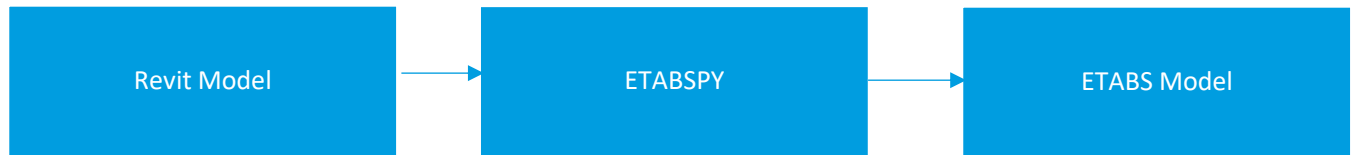
Future versions of ETABSPY may incorporate optimization features to assist in achieving the most economical and code-compliant structural designs. One such feature could involve the use of **gradient descent** algorithms to minimize cost functions—such as material usage, reinforcement quantity, or structural weight—across multiple design variables (e.g., member sizes, reinforcement ratios, wall thicknesses).

By evaluating the gradient of the cost function in an **n-dimensional design space**, the tool could iteratively update design parameters toward an optimal solution. This method would allow engineers to explore efficient solutions within code-defined constraints, reducing overdesign and material waste while maintaining performance and safety.



2. Automatic Model Generation from Revit

Integration with Revit could allow ETABSPY to automatically generate structural models based on architectural BIM data. This would reduce the time spent on manual modeling in ETABS and ensure better coordination between architectural and structural workflows.



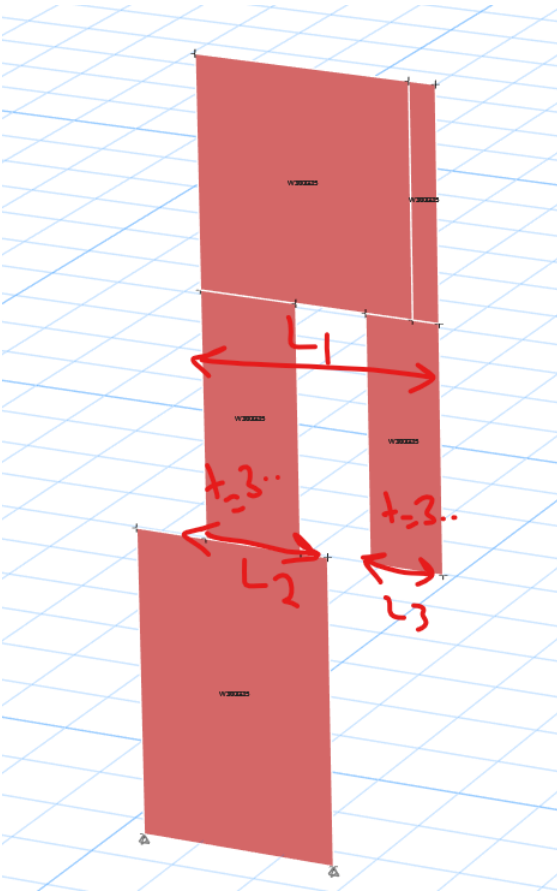


ETABSPY

Error #1:



ETABSPY

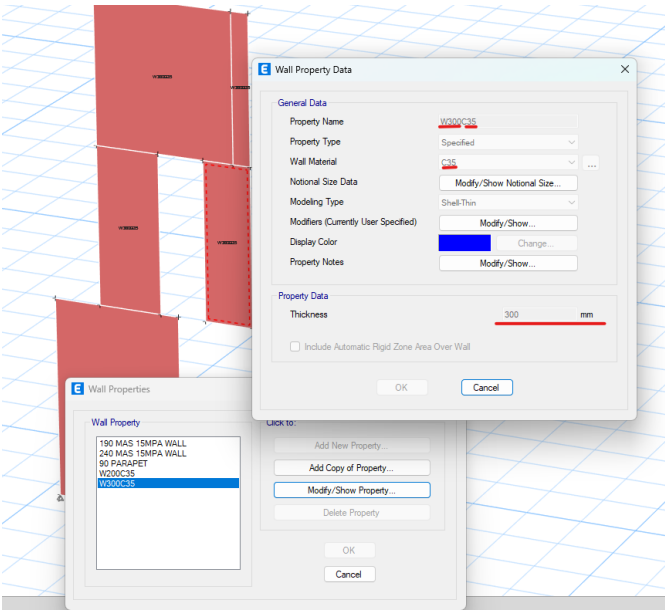
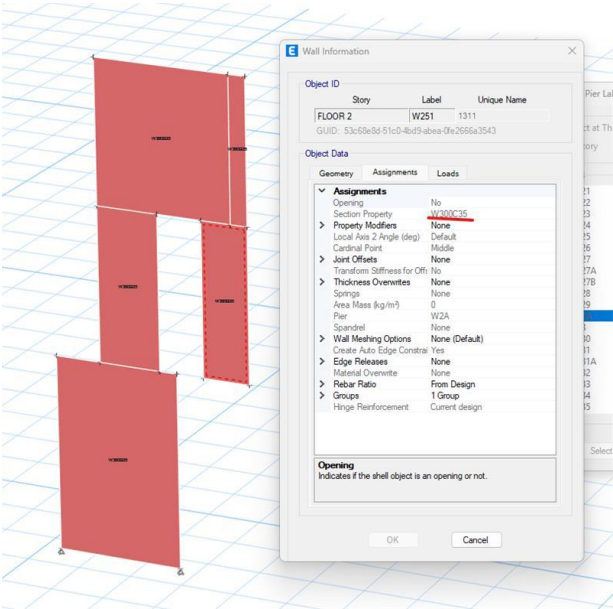


$$t_r = t * \frac{L_2 + L_3}{L_1}$$

10												
11	Story	Axis Angle	No. Area Obj.	No. Line Obj.	Width Bot.	Thk. Bot.	Width Top	Thk. Top	Mat. Prop.	CG Bot X	CG Bot Y	CG Bot Z
12					mm	mm	mm	mm		mm	mm	mm
13												
14	FLOOR 3	0.0	2	0	3600.00	300.00	3600.00	300.00	C35	28260.00	22030.00	8000.00
15	FLOOR 2	0.0	2	0	3600.00	<u>212.50</u>	3600.00	<u>212.50</u>	C35	28187.94	22030.00	4000.00
16	UPPER GR	0.0	1	0	2930.00	300.00	2930.00	300.00	C35	26855.00	22030.00	0.00



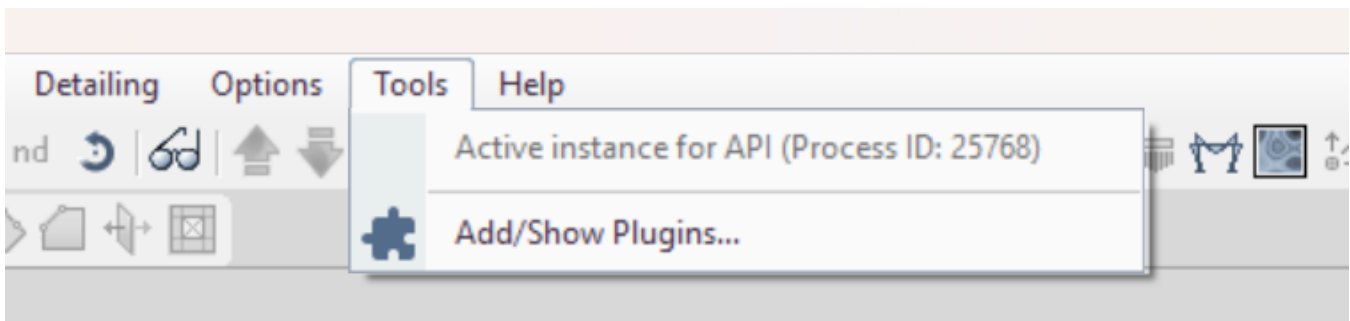
ETABSPY





If faced with this error

```
C:\Coding\ETABSPY>py tests\test_write_sconc_file.py
List of available COM objects.
Attempting to connect to active ETABS instance...
X GetActiveObject failed: [WinError -2147221021] Operation unavailable
X Error: ETABS is not open.
X Script failed...
```



```
Writing SCONCRETE file W5_FLOOR 10_W300_C35.SCO
WARNING: There is no corresponding load combination in combination set 4 for 3.ULS.E-009. Shear force magnification failed.
WARNING: There is no corresponding load combination in combination set 4 for 3.ULS.E-010. Shear force magnification failed.
WARNING: There is no corresponding load combination in combination set 4 for 3.ULS.E-011. Shear force magnification failed.
WARNING: There is no corresponding load combination in combination set 4 for 3.ULS.E-012. Shear force magnification failed.
WARNING: There is no corresponding load combination in combination set 4 for 3.ULS.E-013. Shear force magnification failed.
WARNING: There is no corresponding load combination in combination set 4 for 3.ULS.E-014. Shear force magnification failed.
WARNING: There is no corresponding load combination in combination set 4 for 3.ULS.E-015. Shear force magnification failed.
WARNING: There is no corresponding load combination in combination set 4 for 3.ULS.E-016. Shear force magnification failed.
```